CSL7610 | ARTIFICIAL INTELLIGENCE ASSIGNMENT

CLASSICAL REASONING AGENT

CSL7610 – Artificial Intelligence Assignment

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1 Objective and system overview

The objective is to build an end-to-end system that receives a knowledge base and a query, converts the input into a logical representation, and predicts whether the query follows from the knowledge base. The final decision is produced by classical reasoning procedures rather than by a generative model.

The implementation selects a reasoning route from the structure of the input. English rule-based inputs are handled by a rule parser and a Horn-clause engine; first-order logic (FOL) inputs are handled by a formula parser and resolution; CounterBench inputs are interpreted as structural causal models and evaluated using counterfactual inference.

2 Knowledge representation

2.1 Rule-based knowledge bases

RuleTaker-style English statements are converted deterministically into literals of the form `Lit(neg, predicate, arguments)`. Facts become ground literals, while rules become implications whose bodies contain one or more literals. The parser supports the five paraphrase patterns used in the course-format data: "When A, we can conclude B", "Given that A, it follows that B", "If A, then B", "Based on A, it is evident that B", and "Assuming A, we can say B". It also supports common RuleTaker constructions involving variables, relations, and negation.

The default rule semantics use negation as failure (closed-world assumption): a negative condition is satisfied when the corresponding positive fact cannot be derived. An open-world variant, in which explicit negative literals are represented separately, is retained for the semantics ablation.

2.2 First-order logic inputs

FOL formulas are parsed from the supplied logical notation. The parser recognises quantifiers, connectives, predicate arguments, function terms, and simple equality expressions. Parsed formulas are transformed into negation normal form, standardised apart, Skolemised, and converted to conjunctive normal form (CNF).

Entailment is checked by refutation: the premises are combined with the negation of the query, and a derived empty clause constitutes a proof of entailment.

The current implementation includes tolerant handling of malformed parentheses in some input formulas. Such repairs are a preprocessing accommodation, not a guarantee that the repaired formula preserves the intended meaning; repaired inputs should therefore be tracked during evaluation.

2.3 CounterBench representation

CounterBench descriptions are translated into a structural causal model. Variables are represented as occurrence predicates. Conjunctive causes become rules with multiple antecedents, while alternative causes are represented as separate rules. Negated conditions and exogenous-variable priors are retained for counterfactual evaluation.

The implementation enumerates exogenous assignments consistent with the observed evidence, applies interventions by replacing the intervened variables' incoming causes, and evaluates the target in the resulting worlds. The reported binary decision uses the implementation's probability threshold of 0.5.

3 Reasoning algorithms

3.1 Forward

Forward is the primary engine for rule-based knowledge bases. It starts from known facts and repeatedly applies rules whose antecedents are satisfied, adding newly derived facts until a fixpoint is reached or the query is established. The implementation uses predicate-dependency relevance pruning, stratification for negation, indexed facts, greedy ordering of rule-body joins, and provenance information for proof traces.

3.2 Backward

Backward starts from the query and searches for rules whose conclusions can establish it. It recursively attempts to prove the required antecedents using unification. The implementation uses standardisation apart, an ancestor-loop check, a depth bound, and negation as failure.

3.3 Resolution for FOL

The FOL prover uses binary resolution with factoring and unification, including an occurs check. A set-of-support strategy is used: the premises form the usable set and clauses from the negated conclusion form the support set. Duplicate clauses, tautologies, and selected subsumed clauses are pruned. Search can use BFS, DFS, uniform-cost search (UCS), or A*. Clause-length, term-depth, and time limits bound the computation.

A successful refutation proves entailment. Importantly, reaching a resource limit or exhausting the implemented search without a refutation is not, in general, a proof that the query is false. The current binary evaluation maps non-proofs to the negative label; this limitation should be considered when interpreting FOL results.

4 Experimental setup

The experiments below use the public dataset splits loaded by the notebook. The reported sample counts equal the labelled pools shown in the experiment output. RuleTaker is evaluated with both forward and backward chaining. FOLIO and CounterBench use the automatically selected route. Accuracy is the fraction of predictions matching the dataset labels. Total inference time is the elapsed time for the evaluated split; milliseconds per query are calculated by dividing that time by the number of evaluated examples.

The results are measurements from the latest notebook run supplied for this report. The machine and software environment were not included in the supplied results, so timing figures should be treated as run-specific rather than hardware-independent benchmarks. The latest run reports 107.9 seconds of combined inference time.

5 Results

Dataset split	n	Engine	Accuracy (%)	Total time (s)	ms/query	Unparsed / total
RuleTaker-train	1,000	Forward	98.30	0.26	0.26	0 / 10,267
RuleTaker-train	1,000	Backward	98.30	0.23	0.23	0 / 10,267
RuleTaker-test	250	Forward	97.60	0.06	0.24	0 / 2,591
RuleTaker-test	250	Backward	97.60	0.06	0.24	0 / 2,591
RuleTaker-OOD	375	Forward	99.47	0.09	0.24	0 / 3,663
RuleTaker-OOD	375	Backward	99.47	0.08	0.21	0 / 3,663
FOLIO-train	1,001	Resolution (auto)	80.32	74.44	74.37	29 / 6,286
FOLIO-validation	203	Resolution (auto)	84.24	32.52	160.20	18 / 1,307
CounterBench-alpha	1,000	Causal (auto)	96.00	0.46	0.46	0 / 1,000
CounterBench-backdoor	200	Causal (auto)	90.00	0.09	0.45	0 / 200

Across the seven dataset splits, the notebook reports 3,715 correct predictions out of 4,029 examples (92.21%), with 107.9 seconds of combined inference time. This is a descriptive aggregate across different

datasets and split types; it includes training splits as well as test/validation/OOD splits, so it should not be presented as a single held-out test accuracy.

5.1 Results by dataset

- **RuleTaker:** accuracy is 98.30% on train, 97.60% on test, and 99.47% on OOD. Forward and backward chaining produced identical predictions on all evaluated examples (zero disagreements). Latest run times were 0.26 s / 0.23 s (train), 0.06 s / 0.06 s (test), and 0.09 s / 0.08 s (OOD) for forward / backward chaining.
- **FOLIO:** accuracy is 80.32% on train and 84.24% on validation. The latest run took 74.44 s and 32.52 s, respectively. Diagnostics report 29 unparsed formula lines out of 6,286 on train and 18 out of 1,307 on validation, affecting 28 and 9 examples, respectively.
- **CounterBench:** accuracy is 96.00% on alpha and 90.00% on backdoor. The latest run took 0.46 s and 0.09 s, respectively. The alpha subset reports 100% on basic, 92% on conditional, 96% on joint, and 96% on nested examples.

5.2 Resolution search-strategy comparison

Strategy	FOLIO validation accuracy (%)	Total time (s)
BFS	83.25	41.2
DFS	84.73	58.3
UCS	84.73	35.2
A*	84.24	41.4

In this run, DFS and UCS achieved the highest reported validation accuracy (84.73%). UCS had the lowest measured time (35.2 s), narrowly ahead of BFS (41.2 s) and A* (41.4 s). The difference in accuracy is small and should be interpreted in light of the 203-example validation set and the parser/prover limitations.

5.3 Semantics and engine consistency checks

- **RuleTaker-test:** closed-world/negation-as-failure and open-world settings both achieved 97.60% accuracy (250 examples).
- **Differential testing:** forward and backward chaining were compared on 4,000 generated RuleTaker-style knowledge bases. They disagreed on 16 cases overall; none of the 3,918 stratifiable cases disagreed. The remaining disagreements involved negation through recursion, for which the stated closed-world semantics do not provide a well-defined stratified interpretation.
- The public RuleTaker evaluation itself reported zero forward/backward disagreements on train, test, and OOD splits.

6 Reasoning traces

6.1 Rule-based example

Query: Bob is successful. **Prediction:** True

```
successful(bob) [rule:  smart and kind imply successful]
smart(bob) [rule:  young implies smart]
young(bob) [given fact]
kind(bob) [given fact]
```

6.2 Course-format example

6.3 Latest sanity-test trace

Query: Bob is successful. **Prediction:** True

```
successful(bob) [by rule:  If someone is smart and kind, then they are successful.]
smart(bob) [by rule:  If someone is young, then they are smart.]
young(bob) [given fact]
kind(bob) [given fact]
prop_henry_volunteers [rule:  Liam hikes mountains implies Henry volunteers]
```

```
prop_liam_hikes_mountains [rule:  Liam volunteers implies Liam hikes mountains]
prop_liam_volunteers [given fact]
```

6.4 FOL resolution example

Query: Mortal(socrates). **Prediction:** True. The notebook trace reports two premise clauses and one clause from the negated conclusion. Resolution derives the empty clause, establishing that the premises together with the negated query are inconsistent.

6.5 CounterBench counterfactual examples

Example A: "Would Lumbo occur if not Ziklo instead of Ziklo?" **Prediction:** False. The trace reports evidence Ziklo = 1, intervention $do(Ziklo = 0)$, and a target probability of 0.000.

Latest sanity-test demo: the intervention $do(Nuv = 0)$, $do(Splee = 0)$ produced

$$P(Wrox \mid evidence, do) = 1.000.$$

The reported derivation is Wrox ← Yurd ← Zimb ← (Skrim or Druk) ← Skrim ← Plog ← not Druk, where not Druk is handled by negation as failure. This is a sanity-test example, not a measured dataset result.

7 Error analysis and interpretation

The results show that performance differs substantially by representation. RuleTaker has high accuracy and no parser failures in the evaluated samples. CounterBench also has high accuracy, although the backdoor subset is more difficult. FOLIO is the main source of errors and runtime cost: it has the lowest accuracy, non-zero parser diagnostics, and substantially higher inference time.

The saved error examples indicate several possible sources of mismatch: formulas or annotations that are malformed or repaired during parsing, unsupported or incompletely handled logical constructs, differences in predicate names or argument structure, equality handling, resource limits, and possible label/interpretation mismatches. The current results do not establish that all residual errors are dataset label noise; each error would need manual inspection before making that claim.

For CounterBench, the saved errors occur in both directions on the alpha and backdoor subsets. This suggests that counterfactual handling of observed evidence, interventions, and exogenous priors deserves further testing. The current results alone do not establish that the labels are incorrect.

8 Limitations

- The English rule parser is template-based. Sentences outside its supported patterns may not be represented correctly.
- FOLIO performance is limited by formula coverage and the current prover's bounded search. The parser diagnostics show non-zero unparsed inputs.
- Equality is currently treated as an ordinary predicate; full equality reasoning and substitution are not implemented.
- The implementation's binary fallback for an unsuccessful or resource-limited FOL proof can conflate "not proved" with "not entailed." A three-valued internal status (entailed, refuted/not entailed under the chosen semantics, unknown/resource limit) would be safer.
- CounterBench parsing is based on supported textual templates and the current causal-model assumptions. More varied graph structures and evidence/intervention combinations require additional testing.
- Timing results depend on the execution environment, which was not recorded in the supplied results.
- The aggregate 92.21% combines train, test, OOD, and validation splits; it is not a pure held-out evaluation score.

9 Conclusion

The implemented system demonstrates an end-to-end classical reasoning pipeline using forward chaining, backward chaining, unification, resolution, and structural causal reasoning. It performs strongly on the evaluated RuleTaker and CounterBench samples, while FOLIO remains more challenging because of formula

complexity, parser coverage, and bounded resolution search. The next improvements should focus on making FOL parsing and proof-status handling more reliable, auditing CounterBench counterfactual edge cases, and reporting held-out results separately from training-split results.

End of report